

Dimension one and twelve rational components of four-truncated point modules for $\mathcal{B}(\text{Aff}(3, -1), i)$

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Abstract

Let K be an algebraically closed field of characteristic zero with a chosen $i \in K$ satisfying $i^2 = -1$. For the Nichols algebra of the three-element affine rack with constant cocycle i , we prove that the scheme of cyclic graded left modules with Hilbert series $1 + t + \cdots + t^4$ is nonempty and has reduced dimension one. We exhibit twelve distinct geometrically irreducible components, each isomorphic to $\mathbb{P}_K^1$. The upper bound follows from a complete affine-chart cover and 1,428 exact polynomial ideal-membership identities; the lower bound follows from explicit homogeneous families. This is a partial classification: the complete component list, possible additional isolated components, and the exhaustive closed-point description remain unresolved.

1 Statement and context

Let V have basis v_0, v_1, v_2 , and give it the braiding

$$c(v_x \otimes v_y) = i v_{2x-y \bmod 3} \otimes v_x. \quad (1)$$

Write $A = \mathcal{B}(V, c)$. A four-truncated left point module is a graded cyclic left A -module $M = \bigoplus_{j=0}^4 K m_j$, generated by m_0 . Its action has the form

$$v_g m_j = \lambda_{g,j} m_{j+1} \quad (0 \leq j < 4), \quad v_g m_4 = 0.$$

Cyclicity is equivalent to each action vector $\lambda_j = (\lambda_{0,j}, \lambda_{1,j}, \lambda_{2,j})$ being nonzero. Changing the bases of the graded pieces gives independent rescalings of these four vectors. Thus the parameter scheme $\Gamma_4(A)$ lies in $(\mathbb{P}_K^2)^4$. There are four action vectors, despite the five nonzero graded pieces.

Theorem 1. *For every K and chosen i as above, $\Gamma_4(A)$ is nonempty and*

$$\dim \Gamma_4(A)_{\text{red}} = 1.$$

The twelve images A_x, B_x, C_x, D_x , $x = 0, 1, 2$, defined in (7) are distinct geometrically irreducible components of $\Gamma_4(A)_{\text{red}}$, each isomorphic to $\mathbb{P}_K^1$.

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The theorem does not assert that these are all components. In particular, it does not assert equidimensionality, exclude additional isolated components, or determine the nilpotent structure of $\Gamma_4(A)$. The original question asking for all reduced components, their dimensions, and all closed points therefore remains unresolved.

Greenfeld, Vendramin and Venkatesh study point modules of Nichols algebras over non-abelian groups [1]. Their Proposition 6.1 guarantees nonemptiness through Γ_3 for a constant cocycle of order four; Problem 6.2 asks for the root-of-unity and truncation classification for $\text{Aff}(3, -1)$. Their nearby explicit characteristic-zero calculation concerns a primitive cubic root. Theorem 1 determines the dimension of the next truncation for the fourth root and identifies twelve actual components. A bounded comparison with the primary version and searches conducted on September 7, 2026 located no equivalent result; this is not an exhaustive priority assessment. As adjacent computational-algebra context, Zeng's work on a bosonic–fermionic coinvariant basis uses exact standard-basis and rank certificates [2]. That reference is methodological context only, not an input to the theorem.

2 The complete equations in degree four

For $n \geq 2$, let S_n be the quantum symmetrizer: the sum of the *positive* braid lifts of all permutations in $\mathfrak{S}_n$. The Nichols ideal has degree- n part $\ker S_n$. The braid relation follows from the rack identity for $x \triangleright y = 2x - y$ on $\mathbb{F}_3$. Reduced expressions of a permutation give the same positive lift by the braid relations. In particular, no inverse braid operator is used here.

For clarity, put $[abcd] = v_a v_b v_c v_d$. The seven quartic relations are the three pure fourth powers and the four vectors r_1, r_2, r_3, r_4 specified in Table 1. A row with coefficient α and words W contributes $\alpha \sum_{w \in W} [w]$ to the indicated relation. The table therefore fixes the equations without an implicit presentation convention.

Relation	Coefficient	Words (continuation rows retain the coefficient)
r_1	1	0102, 0201, 1012, 1210, 2021, 2120
	$-i$	0110, 0220, 1001, 1221, 2002, 2112
	-1	0121, 0212, 1020, 1202, 2010, 2101
r_2	1	0011, 0022, 1100, 1122, 2200, 2211
	$1 + i$	0110, 0220, 1001, 1221, 2002, 2112
	2	0121, 0212, 1020, 1202, 2010, 2101
r_3	1	0001, 0012, 0020, 0100, 0111, 0122
		0202, 0210, 0221, 1002, 1010, 1021
		1101, 1112, 1120, 1200, 1211, 1222
		2000, 2011, 2022, 2102, 2110, 2121
		2201, 2212, 2220
r_4	1	0002, 0010, 0021, 0101, 0112, 0120
		0200, 0211, 0222, 1000, 1011, 1022
		1102, 1110, 1121, 1201, 1212, 1220
		2001, 2012, 2020, 2100, 2111, 2122
		2202, 2210, 2221

Table 1: The four non-pure quartic relations. Together with $[0000], [1111], [2222]$, these form the full degree-four relation space.

n	$\dim V^{\otimes n}$	$\text{rank } S_n$	$\dim \ker S_n$
2	9	9	0
3	27	27	0
4	81	74	7

Here is a certificate of these rank assertions. Construct S_n on all lexicographically ordered words in $\{0, 1, 2\}^n$, using Gaussian-integer arithmetic. Its reduction under $\mathbb{Z}[i] \rightarrow \mathbb{F}_5$, $i \mapsto 2$, has ranks 9, 27, 74. These are lower bounds for the characteristic-zero ranks, because a minor nonzero after reduction was already nonzero. Each of the seven displayed quartic vectors is killed exactly by S_4 , and their coefficient vectors have rank seven modulo 13, with $i \mapsto 5$. This gives the matching upper bound and proves that they span the whole quartic kernel. The matrix ranks are also checked at 13; no positive-characteristic conclusion is inferred. The accompanying verifier reconstructs the symmetrizers recursively as

$$S_1 = 1, \quad S_n = (1 \otimes S_{n-1})(1 + c_1 + c_1 c_2 + \cdots + c_1 \cdots c_{n-1}), \quad (2)$$

where c_j acts in positions $j, j+1$, and products act rightmost first. This is the positive-lift sum by the usual permutation-coset decomposition.

If $r = \sum_w r_w [w]$ is any of these quartics, its left-module equation is

$$f_r = \sum_{w=(w_0, w_1, w_2, w_3)} r_w \lambda_{w_0, 3} \lambda_{w_1, 2} \lambda_{w_2, 1} \lambda_{w_3, 0} = 0. \quad (3)$$

The reversal is forced by left action: the rightmost letter acts first. There are no quadratic or cubic relations, a quartic acts trivially from each positive starting degree, and every higher-degree relation acts trivially on M . Consequently these seven multilinear equations define exactly $\Gamma_4(A)$ in $(\mathbb{P}^2)^4$; no higher-degree presentation is needed. Three equations are

$$\prod_{j=0}^3 \lambda_{g, j} = 0 \quad (g = 0, 1, 2). \quad (4)$$

3 The dimension upper bound

We first record the commutative-algebra observation used to interpret the finite certificates.

Lemma 2. *Let $J \subset F[u_1, \dots, u_r]$, $r \geq 2$, be an ideal over a field. Suppose nonzero $g_1, \dots, g_m \in J$ have the following property for a degree-compatible monomial order: every pair of variables contains the support of some $\text{LM}(g_k)$. Then $\dim F[u_1, \dots, u_r]/J \leq 1$.*

Proof. Put $M = \max(1, \max_k \deg \text{LM}(g_k))$. Division by the g_k terminates and does not increase total degree. The monomials divisible by none of the $\text{LM}(g_k)$ therefore span the quotient in every degree filtration; a Gröbner basis hypothesis is unnecessary. A surviving monomial cannot have two exponents at least M , since a leading monomial supported on those two variables would divide it. All but at most one exponent are less than M . The number of survivors of total degree at most N is at most $rM^{r-1}(N+1)$, so the filtered dimensions grow as $O(N)$. If the quotient had dimension at least two, Noether normalization would provide two algebraically independent elements, each represented by a polynomial of some bounded degree d . Their monomials of total degree at most $\lfloor N/d \rfloor$ would give quadratically many independent elements in filtration degree N , a contradiction. The zero quotient and a constant leading monomial cause no exception. $\square$

For $p = (p_0, p_1, p_2, p_3) \in \{0, 1, 2\}^4$, normalize $\lambda_{p_j, j} = 1$. These are the 81 standard affine charts. All six permutations of the generator labels are rack automorphisms: they are exactly the affine maps $x \mapsto ux + v$ of $\mathbb{F}_3$, $u \neq 0$. They preserve the braiding and the Nichols ideal. Chart representatives under this action are

$$\begin{aligned} &0000, 0001, 0010, 0011, 0012, 0100, 0101, \\ &0102, 0110, 0111, 0112, 0120, 0121, 0122. \end{aligned}$$

The first orbit has size three, and each other orbit has size six, accounting for all $3 + 13 \cdot 6 = 81$ charts. The chart 0000 is empty: its pure 0-power equation becomes $1 = 0$.

For every other representative choose $a : \{0, 1, 2\} \rightarrow \{0, 1, 2, 3\}$ and set $\lambda_{g, a(g)} = 0$ for $g = 0, 1, 2$. Retain precisely the assignments satisfying $p_{a(g)} \neq g$ for each g , so that no normalized coordinate is set to zero. There are 204 retained branches. Each has four distinct normalized coordinates, three distinct zero coordinates, and five free coordinates. Every point of the chart belongs to a retained branch by (4). The branches form a finite closed cover of the reduced chart over algebraically closed K . Branches may overlap; coordinate boundary points are included.

On a branch, let $f_1, \dots, f_4$ be the specializations of the four non-pure quartic equations, in their table order. Each is nonzero. In $\mathbb{Q}(i)[u_1, \dots, u_5]$ the electronic certificate gives finite lists of polynomials and identities

$$g_k = \sum_{\ell=1}^4 h_{k, \ell} f_\ell. \quad (5)$$

It contains five identities on 36 branches, seven on 144 branches, and ten on 24 branches, totaling 1,428 identities. The variable order is obtained from the action coordinates in degree order, and then label order, by deleting normalized and zero coordinates. Leading terms use degree reverse lexicographic order with that variable order.

For every branch the verifier multiplies out all identities in exact $\mathbb{Q}(i)$ arithmetic, recomputes the leading monomials, and verifies that each of the ten pairs of free variables contains a recorded leading support. The maximum recorded leading total degree is nine. The verifier also enumerates the full chart and zero-assignment domain and compares it with the certificate, so this is a complete finite cover calculation, not sampling. Section 5 describes the data and reproducible check.

All identities and their nonzero leading coefficients remain valid under the embedding $\mathbb{Q}(i) \hookrightarrow K$ determined by the chosen root. Lemma 2 bounds every branch dimension by one over K itself. Taking the finite cover of each chart, and then all charts, gives

$$\dim \Gamma_4(A)_{\text{red}} \leq 1. \quad (6)$$

4 Twelve projective-line components

Let e_0, e_1, e_2 be the coordinate basis of K^3 , and put $h = e_0 + e_1 + e_2$. For $x \in \{0, 1, 2\}$ let $y < z$ be the other two labels. Define

$$d_x^+ = e_y + e_z, \quad d_x^- = e_y - e_z, \quad f_x = -2e_x + e_y + e_z.$$

$$\begin{aligned} L_x(s, t) &= se_x + t(e_y + e_z), \\ H_x(s, t) &= se_x + te_y - (s + t)e_z, \\ R_x(s, t) &= se_y + te_z. \end{aligned}$$

For $[s : t] \in \mathbb{P}_K^1$, the twelve maps, in increasing action-degree order, are

$$\begin{aligned} A_x([s : t]) &= ([e_x], [L_x(s, t)], [h], [d_x^-]), \\ B_x([s : t]) &= ([e_x], [H_x(s, t)], [h], [d_x^+]), \\ C_x([s : t]) &= ([e_x], [f_x], [h], [R_x(s, t)]), \\ D_x([s : t]) &= ([e_x], [h], [L_x(s, t)], [d_x^-]). \end{aligned} \tag{7}$$

We use the same symbols for their images.

Substitution of each tuple into (3) gives zero for all seven quartics: these are 84 identities in $\mathbb{Q}(i)[s, t]$, checked symbolically in the accompanying verifier. Homogeneous symbolic substitution includes $s = 0$ and $t = 0$. Every fixed block is nonzero in characteristic zero, and each moving block is a rank-two linear map $K^2 \rightarrow K^3$. Thus each tuple defines a morphism on the whole projective line and its moving block is a closed embedding into $\mathbb{P}^2$. Together with the fixed blocks this is a closed embedding into $(\mathbb{P}^2)^4$, with image in $\Gamma_4(A)$. Since $\mathbb{P}^1$ is reduced, the morphism factors through $\Gamma_4(A)_{\text{red}}$. Each image is geometrically irreducible of dimension one, proving the lower bound matching (6).

To distinguish the images, different x give different first blocks. For fixed x , only C_x has a varying last block. The last block of B_x is $[d_x^+]$, different from $[d_x^-]$ in characteristic zero. Finally A_x varies in the second block and is fixed in the third, whereas D_x has the opposite behavior. Hence the twelve images are distinct. Intersections at individual points do not affect this argument.

Each image is a closed irreducible curve and is contained in an irreducible component of $\Gamma_4(A)_{\text{red}}$. The upper bound forces that component to have dimension one. A proper closed irreducible subset of an irreducible finite-type curve has dimension zero, so the image equals the component. The identities and embeddings persist over every algebraically closed extension, proving the geometric assertion. This completes the proof of Theorem 1.

5 Exact certificates and limitations

The accompanying `supplement/` directory contains the complete quartic coefficient vectors in `low_degree_relations.json`, all 204 branches and 1,428 identities in `dimension_certificates.json`, and the homogeneous family data in `rational_families.json`. Its `verify.py` uses only the Python standard library and exact pairs of rational numbers for $\mathbb{Q}(i)$. Run from the manuscript directory:

```
python3 supplement/verify.py
```

The program checks the braid relation on all 27 triples, reconstructs the degree-two through degree-four symmetrizers by (2), checks the exact kernels and modular ranks, regenerates the complete chart cover, verifies every identity and leading-support condition, and substitutes all twelve families. It checks rank two of each moving block, including the absence of projective base points. This is a finite exact certificate for the computations used above, not a substitute for the dimension and maximality arguments.

The discovery calculation used Singular 4.4.1 for standard bases and lift matrices; the original symbolic verifier used SymPy 1.14.0. A separately implemented verifier used rational-pair sparse arithmetic without Singular, SymPy, or imports of the discovery code. The portable verifier supplied here adapts that implementation's file handling; its mathematical checking routines are unchanged. No floating-point fit or random sample enters the proof.

Computational and AI assistance. An OpenAI Codex assistant was used for symbolic research, generation of code and certificates, and preparation of this text. A separate model context reconstructed and checked the argument and exact certificates. That review is neither external peer review nor formal proof-assistant certification. The mathematical argument and its finite verification scope are stated above so they can be assessed directly.

Remaining classification. The theorem establishes the global reduced dimension and the twelve specified components. A complete determination of the remaining components, including any isolated points, requires further work; the formulas in (7) are not an exhaustive closed-point list. No claim about the nonreduced structure is made.

References

- [1] B. Greenfeld, L. Vendramin, and S. Venkatesh. Point modules of Nichols algebras over non-abelian groups. arXiv:2608.03595, version 1, 2026. <https://arxiv.org/abs/2608.03595>.
- [2] Zijian Zeng. The $n = 5$ case of Lentfer’s $(1, 2)$ -bosonic-fermionic coinvariant basis conjecture. SSRN working paper 7379218, 2026. <https://doi.org/10.2139/ssrn.7379218>.